

Contact: ✉ raul.vazquez@helsinki.fi in [linkedin.com/in/jraulvc](https://www.linkedin.com/in/jraulvc)  [jrvz.github.io/](https://github.com/jrvz)

References: 1) Jörg Tiedemann ✉ jorg.tiedemann@helsinki.fi ☎ +358 50 311 9440
2) Mathias Creutz ✉ mathias.creutz@helsinki.fi ☎ +358 40 590 0117
3) Timothee Mickus ✉ timothee.mickus@helsinki.fi ☎ +358 50 479 4177

Education

- 2018–2023 **PhD Language Technology, University of Helsinki (FI).**
Funding from the European Research Council (ERC).
Supervisor: Prof. Jörg Tiedemann & co-Supervisor: Mathias Creutz
Focus Areas: NLP, NMT, Representation Learning
- 2015–2017 **MSc Mathematical Engineering, Università degli Studi dell'Aquila (IT).**
Scholarship award from the European Commission.
Thesis: The Vehicle Routing Problem - An Application and Proposed Heuristics.
Adviser: Prof. Claudio Arbib & co-Adviser: Prof. Stefano Smriglio
Focus Areas: Mathematical Modelling, ODEs, PDEs, Optimization GPA: 110/110
- 2016 **MSc Technomathematik, Hamburg Universität (DE).**
Joint degree awarded as participant of the Erasmus Mundus Joint MSc. in Mathematical Modelling in Engineering - MathMods (<http://www.mathmods.eu/>)
Focus Areas: non-smooth Convex Optimization and Numerical Approximation methods for PDEs (Galerkin Methods, Finite Differences and Finite Volumes) GPA: 2.03/5
- 2008–2014 **Licentiate(BSc+MSc) Applied Mathematics, ITAM (MX).**
Thesis: A Bayesian Approach to Time Series Smoothing via the Hodrick-Prescott Filter.
Adviser: Prof. Enrique de Alba Guerra
Focus areas: Statistics, Probability, Linear and non-Linear Optimization GPA: 8,4/10
- 2011–2012 **Exchange Program from the German Academic Exchange Service (DAAD), Universität des Saarlandes (DE), DAAD Scholarship Award.**

Work Experience

- 2024–NOW **Researcher & Teaching Staff, University of Helsinki** Helsinki (FI).
Research on generative NLP and uncertainty propagation, and Teaching NLP courses
- 2018–2023 **Researcher - Doctoral Candidate, University of Helsinki** Helsinki (FI).
Research on modular & multilingual machine translation.
- 2017–2018 **Data Science Lab partner, Secretariat of Social Development** Mexico/Remote.
Development of an agricultural production monitor via remote-sensing data.
- 2017 **Collaborator, LFondry & Università degli Studi di L'Aquila** L'Aquila (IT).
Customization of the Vehicle Routing Problem for improving LFondry's production line.
- 2014–2015 **Project Coordinator, INEGI - Mexico's official statistics agency** Mexico (MX).
Assisting the INEGI's vice-president. Research on time series and Bayesian statistics.
- 2016 **Modelling Camp Participant, Hamburg Universität** Hamburg (DE).
Implementation & simulation of a mathematical framework for describing crowd dynamics.
- 2012 **Vehicle Inspection Intern, Volkswagen** Hannover (DE).
Creation of a GUI to improve the production line operation using an infrared signal manager.

Languages

Native **Spanish**

C2 **English**

Toefl iBT, 2014 (107/120)

C1 **Italian**

DALI-C1, 2017 (80.9%)

C1 **German**

DSH-2, 2012 (86% oral, 67% written)

B2 **Finnish**

YKI-keskitaso, 2020

Teaching Merits

- **Main Teacher**, Intro to Neural Machine Translation University of Helsinki.
5CR Master's course on generative AI, applied to machine translation 2024,2023
- **Co-Teacher**, Intro. to Neural Machine Translation University of Helsinki.
idem. 2021,2022
- **Co-Teacher**, Building NLP Applications University of Helsinki.
5CR Bachelor's course where students create an information retrieval app. 2020,2021
- **Co-Teacher**, Mathematics for Linguists University of Helsinki.
Intro course for Bachelor's students to provide foundations for NLP 2019,2020,2021
- **Teaching Assistant**, Machine learning for Linguists University of Helsinki.
Intro course for Bachelor's students to provide foundations for NLP 2019,2020
- **Trainee**, University Pedagogics University of Helsinki.
Trainee at the pedagogical training program at the university. Progress: 25cr/60cr

Publications

- 2026 **Knowledge Distillation as Decontamination? Revisiting the “Data Laundering” Concern in Classification Tasks**, *H. Luo, R. Vázquez, T. Mickus, F. Ginter, J. Tiedemann. In ICLR 2026 (accepted/in press).*
- Building Multimodal Corpora Using Microtask Pipelines and Local Annotators**, *H. Hotti, R. Vázquez, A.-K. E. Jokipohja, T. Kallioikoski, H. Paakki, R. Suviranta, T. Hiippala. In LREC-COLING 2026.*
- 2025 **SemEval-2025 Task 3: Mu-SHROOM, the Multilingual Shared-task on Hallucinations and Related Observable Overgeneration Mistakes**, *R. Vázquez, T. Mickus, E. Zosa, T. Vahtola, J. Tiedemann, A. Sinha, V. Segonne, F. Sánchez-Vega, A. Raganato, J. Libovický, J. Karlgren, S. Ji, J. Helcl, L. Guillou, O. de Gibert, J. Bengoetxea, J. Attieh, M. Apidianaki. In SemEval 2025.*
- Your Model is Overconfident, and Other Lies We Tell Ourselves**, *T. Mickus, A. Sinha, R. Vázquez. In ACL 2025.*
- Scaling Low-Resource MT via Synthetic Data Generation with LLMs**, *O. de Gibert, J. Attieh, T. Vahtola, M. Aulamo, Z. Li, R. Vázquez, T. Hu, J. Tiedemann. In EMNLP 2025.*
- Proceedings of the 2nd Workshop on Uncertainty-Aware NLP (UncertainNLP 2025)**, *B. Eikema, R. Vázquez, J. Berant, M.-C. de Marneffe, B. Plank, A. Shelmanov, S. Swayamdipta, J. Tiedemann, C. Zerva, W. Aziz (Editors).*

- GlottEval: A Test Suite for Massively Multilingual Evaluation of Large Language Models**, H. Luo, Z. Li, J. Attieh, S. Devkota, O. de Gibert, X. Huang, S. Ji, P. Lin, B. S. P. V. Mantina, A. Sreenidhi, R. Vázquez, M. Wang, S. Yusofi, F. Yuan, J. Tiedemann. In *EMNLP 2025 (System Demonstrations)*.
- Findings of the AmericasNLP 2025 Shared Tasks on Machine Translation, Creation of Educational Material, and Translation Metrics for Indigenous Languages of the Americas**, O. de Gibert, R. Pugh, A. Marashian, R. Vázquez, A. Ebrahimi et al. In *AmericasNLP 2025*.
- 2024 **SemEval-2024 Task 6: SHROOM, a Shared-task on Hallucinations and Related Observable Overgeneration Mistakes** *Proceedings of SemEval-2024*, T. Mickus, E. Zosa, R. Vázquez, T. Vahtola, J. Tiedemann, V. Segonne, A. Raganato, M. Apidianaki. In *SemEval 2024*.
- Proceedings of the 1st Workshop on Uncertainty-Aware NLP (UncertainNLP)**, R. Vázquez, H. Celikkanat, D. Ulmer, J. Tiedemann, S. Swayamdipta, W. Aziz, B. Plank, J. Baan, M. de Marneffe. In *EACL 2024 (UncertainNLP)*.
- Proceedings of the 1st Workshop on Modular and Open Multilingual NLP (MOOMIN)**, R. Vázquez, T. Mickus, J. Tiedemann, I. Vulić, A. Üstün. In *EACL 2024 (MOOMIN)*.
- I Have an Attention Bridge to Sell You: Generalization Capabilities of Modular Translation Architectures**, T. Mickus, R. Vázquez, J. Attieh. In *Workshop on Insights from Negative Results in NLP*.
- MAMMOTH: Massively Multilingual Modular Open Translation @ Helsinki**, T. Mickus, S. Grönroos, J. Attieh, M. Boggia, O. De Gibert, S. Ji, N. Loppi, A. Raganato, R. Vázquez, J. Tiedemann. In *EACL2024*.
- 2023 **Why Bother with Geometry? On the Relevance of Linear Decompositions of Transformer Embeddings**, T. Mickus, R. Vázquez. In *BlackBoxNLP2023*.
- Four Approaches to Low-Resource Multilingual NMT: The Helsinki Submission to the AmericasNLP 2023 Shared Task**, O. De Gibert, R. Vázquez, M. Aulamo, Y. Scherrer, S. Virpioja, J. Tiedemann. In *NAACL2023 (AmericasNLP)*.
- Dozens of Translation Directions or Millions of Shared Parameters? Comparing Two Types of Multilinguality in Modular Machine Translation**, M. Boggia, S.A. Grönroos, N. Loppi, T. Mickus, A. Raganato, J. Tiedemann, R. Vázquez. In *NoDaLiDa2023*.
- 2022 **A Closer Look at Parameter Contributions When Training Neural Language and Translation Models**, R. Vázquez, H. Celikkanat, V. Ravishankar, M. Creutz, J. Tiedemann. *EAMT2022*.
- Latest Development in the FoTran Project – Scaling Up Language Coverage in Neural Machine Translation Using Distributed Training with Language-Specific Components**, R. Vázquez, M. Boggia, A. Raganato, N. A. Loppi, S. Grönroos, J. Tiedemann. In *EAMT2022*.
- 2021 **An Empirical Investigation of Word Alignment Supervision for Zero-shot Multilingual Neural Machine Translation**, A. Raganato, R. Vázquez, M. Creutz and J. Tiedemann. In *EMNLP2021*.
- On the differences between BERT and MT encoder spaces and how to address them in translation tasks**, R. Vázquez, H. Celikkanat, M. Creutz and J. Tiedemann. In *ACL-ICJNLP2021-SRW*.

- The Helsinki Submission to the AmericasNLP shared task.**, R. Vázquez, Y. Scherer, S. Virpioja and J. Tiedemann. In *NAACL2021 (AmericasNLP)* .
- 2020 **A Systematic Study of Inner-Attention-Based Sentence Representations in Multilingual Neural Machine Translation**, R. Vázquez, A. Raganato, J. Tiedemann and M. Creutz. In *C. Linguistics. Volume 46, Issue 2.*
- The University of Helsinki Submission to the IWSLT2020 Offline Speech Translation Task.**, R. Vázquez, M. Aulamo, U. Sulubacak, and J. Tiedemann. In *IWSLT2020*.
- 2019 **Multilingual NMT with a language-independent attention bridge**, R. Vázquez, A. Raganato, J. Tiedemann and M. Creutz. In *RepL4NLP2019.*
- An Evaluation of Language-Agnostic Inner-Attention-Based Representations in Machine Translation**, A. Raganato, R. Vázquez, M. Creutz and J. Tiedemann. In *RepL4NLP2019.*
- The University of Helsinki Submission to the WMT2019 Parallel Corpus Filtering Task**, R. Vázquez, U. Sulubacak and J. Tiedemann. In *WMT2019*.
- The University of Helsinki Submissions to the WMT2019 Similar Languages Translation Task**, Y. Scherrer, R. Vázquez, and S. Virpioja. In *WMT2019*.
- The University of Helsinki Submission to the WMT2019 News Translation Task**, A. Talman, U. Sulubacak, R. Vázquez, Y. Scherrer, S. Virpioja, A. Raganato, A. Hurskainen, and J. Tiedemann. In *WMT2019*.
- 2018 **The MeMAD Submission to the WMT18 Multimodal Translation Task**, S.A. Grönroos, B. Huet, M. Kurimo, J. Laaksonen, B. Merialdo, P. Pham, M. Sjöberg, U. Sulubacak, J. Tiedemann, R. Troncy and R. Vázquez. In *WMT2018.*